

Christopher Ochoa

McAllen, TX | (956) 451 - 6889 | Chrisochoaa1@gmail.com

EDUCATION

University of Oklahoma | Norman, OK

Gallogly College of Engineering

B.S Aerospace Engineering - GPA: 3.51

M.S Aerospace Engineering

LinkedIn: www.linkedin.com/in/chrisochoaa/

Aug 2021 - May 2026

Expected May 2027

EXPERIENCE

SOONER ADVANCED MANUFACTURING LAB | Graduate Researcher | Norman, OK

2026 – Present

- Support additive manufacturing and DED/LPBF repair research at the Sooner Advanced Manufacturing Lab for aerospace applications
- Design repair and build strategies, and use FEA to evaluate printability, structural performance, and process trade-offs.

UNDERGRADUATE RESEARCH | Undergraduate Researcher | Norman, OK

2024 – 2025

- Conducted funded research on additive manufacturing of regenerative-cooled rocket nozzles.
- Performed CFD and structural simulations to evaluate thermal and mechanical behaviour of lattice-cooled nozzle designs.

CAMPUS & LEADERSHIP INVOLVEMENT

DESIGN BUILD FLY TEAM | Structures Lead | Norman, OK

2024 – Present

- Lead the structures sub team in the design, analysis, and fabrication of a competition RC aircraft.
- Overseeing CAD modeling, FEA, aircraft fabrication, and composite layup process to ensure structural integrity and mission success.

BOOMER ROCKET TEAM | Team Member | Norman, OK

2023 – Present

- Designed, built, and tested low-powered and high-power rockets, incorporating fiberglass layups and predictive performance modeling.

PRESIDENTS COMMUNITY SCHOLARS | Active Member

2021 – Present

- Developed leadership and service skills through mentorship programs, keynote engagements, and community service projects.

SIGMA CHI FRATERNITY | Historian/Philanthropy Chair

2021 – Present

- Served as chapter Historian and Philanthropy Chair, overseeing event documentation and coordinating community fundraising.

OKLAHOMA DUCKS UNLIMITED | Social/Merchandise Chair

2024 – Present

- Support wetland conservation and habitat restoration initiatives while organizing social events and managing chapter merchandise operations.

PROJECTS

ADDITIVELY MANUFACTURED ROCKET NOZZLE

2025

- Designed, analysed, and printed a regenerative-cooled Ti-64 rocket nozzle with BCC lattice-structured cooling channels for additive manufacturing.
- Used FEA and build-strategy trade studies to evaluate thermal performance, structural integrity, and printability of the nozzle design.

EXTRACURRICULAR ACHIEVEMENTS

EAGLE SCOUT | Boy Scout Troop 7

Aug 2017 – Aug 2021

- Lead an Eagle Scout service project rebuilding the interior of a barn for a no-kill animal shelter, coordinating volunteers and managing all project logistics.

INTERNATIONAL BACCALAUREATE (IB) DIPLOMA

Aug 2017 – May 2021

- Awarded the International Baccalaureate (IB) Diploma for completing a globally, recognized, rigorous academic program.

SKILLS & HOBBIES

Design/Software: SolidWorks, ANSYS, C++, Python, Pan Optimization FEA

Manufacturing: MIG welding, CNC milling, CNC plasma cutting, EDM, Ge Concept Laser M2 LPBF, GE Concept Laser MLabs LPBF

Hobbies: Offshore/Inshore Fishing, Camping, Backpacking, CrossFit

Languages: English: Professional Proficiency

Spanish: Working Proficiency